Hybrid Blockchain-Based Voting System with Machine Learning-Based Fraud Detection.

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Abstract

Traditional voting methods and many existing electronic voting systems face significant challenges such as identity fraud, duplicate voting, vote tampering, delayed result processing, and lack of transparency, which undermine public trust in digital elections. This paper proposes a hybrid e-voting framework that integrates machine learning, biometric authentication, and blockchain technology to provide a secure, transparent, and reliable solution. The system employs a Convolutional Neural Network (CNN) based facial recognition model as a biometric gatekeeper to authenticate voters by calculating a confidence probability, ensuring robust identity verification and minimizing false alarms through costsensitive learning. Suspicious activities like low-confidence matches, multiple voting attempts, and unauthorized access like trigger security alerts, while a validation layer prevents duplicate voting. Once authenticated, votes are securely stored on a blockchain, where each vote is hashed and linked to previous blocks to ensure immutability and detect tampering. The system continuously monitors the hash chain integrity, sending warnings if any unauthorized modifications occur. Additional security measures such as rate limiting and session locking further protect against bot attacks. This framework demonstrates a practical application of machine learning and blockchain in e-voting, addressing critiques on AI usage and blockchain effectiveness by emphasizing biometric probability-based authentication and blockchain's essential role in vote integrity.

Keywords: *Hybrid E-voting, Facial Recognition, Machine Learning, Blockchain Security, Biometric Authentication*

Introduction

Blockchain-based voting systems have emerged as a compelling alternative to conventional electoral mechanisms by recording each vote in an immutable and verifiable ledger. This approach reduces risks of tampering, duplication, and fraud while enhancing transparency throughout the voting process. The decentralized nature of blockchain, reinforced by cryptographic validation, ensures that no single authority can alter voting records, thereby significantly increasing trust and reliability in digital elections. As highlighted in [1], these